AFTER THE MIDTERM: TEST → DELIVER

Post-midterm plan

From here on, every team **tests its system with real users** and then **presents the final project**. The schedule has changed in several places: **an online class, a cancelled class, and a Saturday session**. Please read every slide.

14 Oct – 18 Nov 2026

Test Plan, Guest, Testing, Final

Dr. Prasara Jakkaew, ADT MFU

TIMELINE AT A GLANCE

The new schedule on one slide

Wed 14 Oct
W9

ONLINE

Online class at the usual time. Topic: Testing & UAT, Testing Pyramid

Wed 21 Oct
W10

NO CLASS

No class, replaced by the Saturday guest lecture on 31 Oct

Before Wed 28 Oct
Deadline

DUE

Submit your Test Plan, one per team

Sat 31 Oct
09:00–17:00

GUEST

Cybersecurity guest lecture, onsite in Room E4614; bring a laptop

Wed 4 Nov
W12

TESTING

No class in the room; teams run their tests following the Test Plan

Wed 11 / 18 Nov
W13, W14

FINAL

Final project presentations: teams 1–10 on the 11th, teams 11–20 on the 18th

22 Nov – 4 Dec

Exam period

EXAM

Final exam, emphasis on testing, AI governance, IT/AI law

W9: ONLINE CLASS

14 Oct 2026, Wednesday: online class

HOW TO JOIN

Don't come to the room

- Join online **at the usual class time**
- Zoom: NewGen Zoom room
  Meeting ID 953 5346 9386, Passcode 960242
- **Attendance is taken as usual**; join on time
- Have your mic and internet ready

TOPIC

Testing & UAT

- **Testing Pyramid**: Unit → Integration → E2E
- **UAT**: real users decide whether the system is acceptable
- Today's content is the foundation of the **Test Plan** you must submit

IN PLAIN TERMS This is a normal class, **just online**. Missing it counts as an absence, and you will use today's content for your Test Plan right away.

W10: NO CLASS

21 Oct 2026, Wednesday: no class

WHY

Replaced by Saturday 31 Oct

This week's class hours move to a **full-day guest lecture** on Saturday 31 Oct (see the guest lecture slide).

USE THE TIME

Work on your Test Plan

- Meet as a team and split the Test Plan writing
- Book your **real users** in advance for testing on 4 Nov
- The Test Plan is due **before Wednesday 28 Oct**

DELIVERABLE: DUE BEFORE WED 28 OCT 2026

Test Plan: what it must include

01 Scope

What you will test

Your **MUST** core workflow and its related requirements. Also state what you will **not** test.

02 Levels

Which test levels

Unit, Integration and E2E following the **Testing Pyramid**, plus **UAT** with real users.

03 Test cases

Test cases and UAT scenarios

Steps, test data, **expected result**, and pass/fail for each case.

04 Users

Real users

Who you will test with, how many people, and how many sessions (target **≥ 10 sessions**).

05 Metrics

How you measure

Task results (**before / after**) are **required**. **SUS**, **TAM** or **UTAUT2** are **optional**; if you skip them, describe your own way of measuring.

06 Schedule

Who does what, when

Test schedule (testing on **4 Nov**), owners, and exit criteria.

AI-FIRST You may use AI to draft test cases, but **your team must review every case** and state in the Test Plan where you used AI.

Submit via: `[add submission channel]`, one per team

GUEST LECTURE: SATURDAY, ONSITE

31 Oct 2026, Saturday, 09:00–17:00

Cybersecurity for Software & Systems Engineers

Guest speaker: **Mr. Jedsada Punyapreedapong**, Army Cyber Center

SESSIONS

- **From Code to Breach**
- **AppSec & API Security**
- **Hands-On Lab "Hack & Patch"**
- **DevSecOps & Supply Chain**
- **Defensive Architecture & Observability**
- **Career & Q&A**

BRING & KNOW

- **Room E4614**, Building E4
- **Bring a laptop** for the lab
- **Counts as class time**; attendance is taken
- Complete the **feedback survey** afterwards

W12: NO CLASS, FIELD TESTING

4 Nov 2026, Wednesday: no class in the room

DO

Test with real users

- Follow the **Test Plan** you submitted
- Have real users complete the core workflow **end to end**
- Record every session: pass/fail and issues found
- Fix the critical bugs from feedback

KEEP EVIDENCE

Evidence you need for the final

- **User Evidence Log** (P3): who you tested with, when, and what you found
- **Impact Metrics** (P4): before / after numbers
- Photos or video of testing (ask users for permission first)

AI POLICY Do **not fabricate test results or user interviews**. Every session must really happen, with real people.

HOW TO RUN A TEST SESSION (~30 MIN PER USER)

Testing with real users: one session, step by step

Before

Prepare

- Recruit users from your **real target group**, not classmates
- Get **informed consent (PDPA)** and ask before recording audio or video
- Use codes **U01, U02…** instead of names
- Prepare tasks, questionnaires, and an Excel sheet (one row per user)
- Run **one pilot** first

During

Observe, don't help

- Split roles: a **Moderator** (talks and gives tasks) and a **Note-taker** (times and records)
- Tell the user "we are testing the system, not you"
- Ask them to **think aloud**
- Give tasks **one at a time** and **never help or point**
- Record success, time, where they got stuck, and key quotes

After

Measure right away

- If you use the **SUS**, give it immediately after the tasks
- If you use **TAM or UTAUT2**, give it after the SUS
- A 3-question interview: what did you like, where did you get stuck, would you keep using it and why
- Thank the user and **enter results into the sheet the same day**

Intro + consent **3'** → Old way (baseline) **5'** → **Tasks + think-aloud 10–15'** →
Questionnaires (if used) **3–6'** → Interview **5'**

TIP About 5 users per round usually uncover most of the major usability problems. If you can, **test 5 → fix → test 5 more**: you reach ≥ 10 sessions and get before/after-fix evidence too. (Nielsen, 2000)

TASK-BASED TESTING

Tasks: what users try to do

HOW TO WRITE TASKS

- 3–5 tasks drawn from your locked **core workflow (MUST)**
- Write them as **scenarios**, not button instructions
- Each task has a clear **success criterion**
- Order from easy to hard and use the **same set for everyone**
- Measure: **success**, **time on task**, and **errors**
- Have users do it **the old way** first (e.g., on paper) to get the **before** time

EXAMPLE: CUPLOG

- **T1** "You just sold one latte. Record it."
  Success = correct item logged in ≤ 3 taps
- **T2** "The owner asks how many cups you sold today."
  Success = correct number given
- **T3** "You logged an Americano by mistake. Fix it."
  Success = entry corrected to latte

THE TRAP **Wrong:** "Tap + and choose Latte." (You gave away the answer, so the result means nothing.)
Right: "You just sold one latte. Record it." (The user has to find the way.)

MEASUREMENT: REQUIRED AND OPTIONAL

What is required and what is optional

Required: Tasks (Performance)

"Does it actually work?"

Success rate, time per task, errors, and **before / after** time. This is your evidence for **P4 Impact**.

Optional: SUS (Usability)

"How easy is it to use?"

A **0–100** score per user, averaged and compared with the benchmark of **68**.

Optional: TAM / UTAUT2 (Acceptance)

"Will they adopt it?"

Mean score per construct, such as usefulness, ease of use, and intention to keep using.

HOW TO CHOOSE Use SUS, TAM or UTAUT2: **one, several, or none** (pick only one of TAM and UTAUT2). Combining them gives a clearer picture: a system can be **easy to use** (high SUS) while users **see no benefit** (low PU), so they won't keep using it.
If you use **none** of them, you must present other evidence; see the next slide.

NO SUS, TAM OR UTAUT2? THEN SHOW YOUR EVIDENCE

Not using a standard instrument: what evidence to present

1 Method (What)

Which method you used

Name your method, e.g., task-based observation, post-use interviews, a multi-day field trial with usage logs, or a questionnaire your team wrote (attach the questions), and **why you chose it**.

2 Procedure (How)

How you ran it

Who, how many people, how many sessions, when and where; what users did, what you measured, and who in your team did what.

3 Results

What you found

Measured numbers such as success rate, **before / after** time, or number of uses; problems found with severity, user quotes, and **what your team fixed**.

4 Evidence

Traceable proof

Consent forms, photos or video, the raw data sheet, interview notes, or usage logs that **the instructor can verify**.

THE BAR Whatever method you choose, **you must test with real users** and be able to say **whether the system really helps them, and how you know**.
Vague claims such as "users said it was good", with no method and no evidence, do not count as test results.

OPTIONAL: SYSTEM USABILITY SCALE (BROOKE, 1996)

SUS: 10 items, a 0–100 score

10 ITEMS, LIKERT 1–5 (STRONGLY DISAGREE → STRONGLY AGREE)

1. I think that I would like to use this system frequently.
2. I found the system unnecessarily complex.
3. I thought the system was easy to use.
4. I think that I would need the support of a technical person to be able to use this system.
5. I found the various functions in this system were well integrated.
6. I thought there was too much inconsistency in this system.
7. I would imagine that most people would learn to use this system very quickly.
8. I found the system very cumbersome to use.
9. I felt very confident using the system.
10. I needed to learn a lot of things before I could get going with this system.

SCORING

- Odd items (1, 3, 5, 7, 9): **score − 1**
- Even items (2, 4, 6, 8, 10): **5 − score**
- Sum all 10 **× 2.5** → 0–100 per user
- Report the mean and SD across users

INTERPRETATION (SAURO & LEWIS)

- **≥ 80.8** = A (excellent)
- **68** = average (C)
- **< 51.7** = F (needs fixing)

WATCH OUT SUS is **not a percentage** (78 ≠ 78%)
don't drop or reword items (you may replace "system" with your product name)
use the overall score, not item-by-item
about 12–14 users give fairly stable results (Tullis & Stetson, 2004)

OPTIONAL: TECHNOLOGY ACCEPTANCE (IF USED, PICK ONE MODEL)

TAM or UTAUT2: will users accept it?

TAM (DAVIS, 1989): FITS WORKPLACE TOOLS

- **PU** Perceived Usefulness
  "Using [system] helps me do ___ faster."
- **PEOU** Perceived Ease of Use
  "Learning to use [system] is easy for me."
- **BI** Behavioral Intention
  "I intend to keep using [system]."
- **3–4 items** per construct

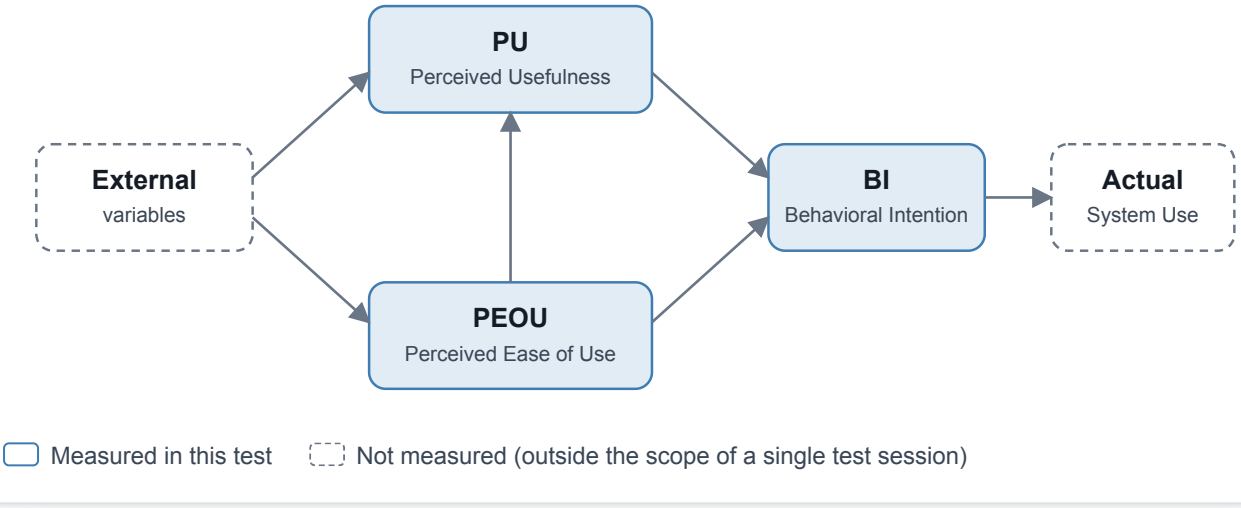
UTAUT2 (VENKATESH, THONG & XU, 2012): FITS CONSUMER APPS

- **PE** Performance Expectancy, **EE** Effort Expectancy
- **SI** Social Influence, **FC** Facilitating Conditions
- **HM** Hedonic Motivation, **PV** Price Value, **HT** Habit
- All point to **BI** (intention to use)
- Pick only the constructs that fit your system
- **HT** can't be measured in a single session; use **PV** only for paid products

RIGOR If you use one, choose **one model** and justify it in your Test Plan
use a 5-point Likert scale
with ~10–15 users, **report only the mean ± SD per construct**
Don't claim causal paths such as "PU drives BI": that needs SEM/PLS and a much larger sample.

MODEL: TAM (DAVIS, 1989; VENKATESH & DAVIS, 1996)

Technology Acceptance Model



PEOU → PU

A system that is **easy to use** makes its **usefulness** easier to see.

PU → BI

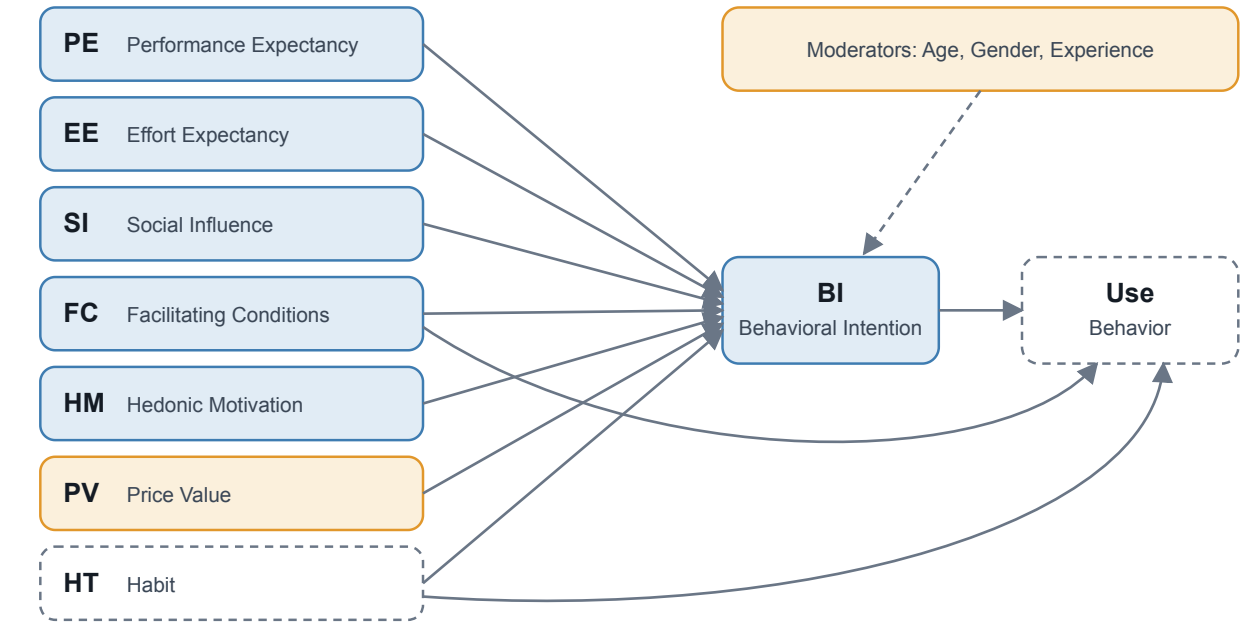
When users see that the system **really helps their work**, they **intend to keep using it**. In many studies this is the strongest path.

PEOU → BI

Ease of use affects intention both **directly** and **through PU**.

MODEL: UTAUT2 (VENKATESH, THONG & XU, 2012)

Unified Theory of Acceptance and Use of Technology 2



Measurable in this test

Measure when relevant (PV only for paid products; moderators collected as participant background)

Cannot be measured in a single session

READING THE MODEL All seven constructs point to **BI**. **FC** and **HT** also affect actual use (Use Behavior) directly. **Age, gender and experience** make these relationships stronger or weaker for different groups.

OPTIONAL: HOW TO USE THE MODEL IN YOUR PROJECT

Using a model in your project: 7 steps

1 **Pick the model that fits the context**

A workplace system that people are required to use calls for **TAM**. A consumer app that people choose for themselves calls for **UTAUT2**. Write your reason in the Test Plan.

2 **Choose the constructs to measure**

For TAM, measure all three: **PU, PEOU, BI**. For UTAUT2, pick the constructs that fit your system and always include **BI**. Leave out HT and Use Behavior.

3 **Build items from the source papers**

Adapt the items in the original papers, **3–4 per construct**, changing only the system and task names. Don't write new items from scratch.

4 **Translate and check meaning**

Translate into Thai, have someone else translate it back into English (**back-translation**), and compare the meaning. Then pilot it with 2–3 people.

5 **Collect the data**

Give the questionnaire **after the tasks and the SUS**, on a 5-point Likert scale. With UTAUT2, also record age, gender, and experience.

6 **Score it**

Each user's construct score = **the mean of that construct's items**. Then report the mean and SD across users.

7 **Interpret and fix**

Link the **lowest-scoring** construct to problems you observed in the tasks, then fix the system. Don't claim causal relationships.

WRITING QUESTIONNAIRE ITEMS

How to write good questions

RULES

- Start from **validated items** in the source papers; change only the system and task names
- **One idea per item**: never ask two things at once
- Name the **system and task** clearly, e.g., "record sales", not "do my work"
- No **leading wording** such as "this excellent system"
- Avoid **negative** and double-negative statements
- Use **the same scale for every item**: 1 = strongly disagree to 5 = strongly agree
- Give **every item a code** such as PU1, PU2, PU3 for analysis

EXAMPLES: WRONG AND RIGHT

- **Wrong:** "CupLog is easy and fast." (two ideas)
  Right: split into "CupLog is easy to use." (PEOU) and "Using CupLog lets me record sales more quickly." (PU)
- **Wrong:** "Don't you like this system that makes work easier?" (leading)
  Right: "I find CupLog useful in my work at the café."
- **Wrong:** "I don't think CupLog is hard to use." (double negative)
  Right: "CupLog is easy to use."
- **Wrong:** "This system is good." (too broad; which construct?)
  Right: word each item to match the construct it measures

ITEM BANK: TAM FOR CUPLOG (LIKERT 1–5)

Example TAM questionnaire

CODE	CONSTRUCT	ITEM
PU1	Perceived Usefulness	Using CupLog lets me record sales more quickly.
PU2	Perceived Usefulness	Using CupLog makes it easier to know the day's sales.
PU3	Perceived Usefulness	I find CupLog useful in my work at the café.
PEOU1	Perceived Ease of Use	Learning to use CupLog is easy for me.
PEOU2	Perceived Ease of Use	My interaction with CupLog is clear and understandable.
PEOU3	Perceived Ease of Use	I find CupLog easy to use.
BI1	Behavioral Intention	I intend to use CupLog in my work.
BI2	Behavioral Intention	I plan to keep using CupLog next month.
BI3	Behavioral Intention	If CupLog were available, I would use it regularly.

SOURCE Adapted from Davis (1989) and Venkatesh & Davis (2000). Replace "CupLog" and "record sales" with your own system and its core task.

ITEM TEMPLATE: UTAUT2 FOR A CONSUMER APP (LIKERT 1–5)

Example UTAUT2 items

CODE	CONSTRUCT	EXAMPLE ITEM
PE1	Performance Expectancy	I find [app] useful in my daily life.
EE1	Effort Expectancy	Learning how to use [app] is easy for me.
SI1	Social Influence	People who are important to me think that I should use [app].
FC1	Facilitating Conditions	I have the knowledge necessary to use [app].
HM1	Hedonic Motivation	Using [app] is fun.
PV1	Price Value	[app] is good value for the money.
BI1	Behavioral Intention	I intend to continue using [app] in the future.

NOTE This table shows **one example per construct**. Your real questionnaire needs **3–4 items per construct**, adapted from Venkatesh, Thong & Xu (2012), and only for the constructs that fit your system.

SCORING: FROM ANSWERS TO CONSTRUCT SCORES

Scoring constructs in Excel

ILLUSTRATIVE DATA: PU

USER	PU1	PU2	PU3	PU (MEAN)
U01	4	5	4	**4.33**
U02	5	5	4	**4.67**
U03	3	4	4	**3.67**
PU across users (mean ± SD)				**4.22 ± 0.51**

Do the same for every construct, then chart the construct means as horizontal bars.

EXCEL FORMULAS (ONE ROW PER USER)

```
' PU score for the user in row 2 (PU1–PU3 in B:D)
=AVERAGE(B2:D2)

' Mean and SD across users (PU scores in column E)
=AVERAGE(E2:E11)
=STDEV.S(E2:E11)

' SUS for the user in row 2 (items 1–10 in B:K)
=2.5*((B2-1)+(5-C2)+(D2-1)+(5-E2)+(F2-1)
     +(5-G2)+(H2-1)+(5-I2)+(J2-1)+(5-K2))
```

OPTIONAL You may compute **Cronbach's α** per construct (≥ 0.70 is acceptable), but with 10–15 users the value is unstable, so report it as supporting information only.

DELIVERABLES FROM TESTING → FINAL

What to bring back and present in the final

#	WHAT TO PRESENT	HOW TO SHOW IT	PORTFOLIO
1	**Participants**: number of users and sessions, user groups (anonymized)	Table U01–U10	P3
2	**Task results**: success rate, mean time, errors, before vs after time	Table + before / after chart	P4
3	**SUS** (if used): mean, SD, grade vs 68	Bar per user + line at 68	P4
4	**TAM / UTAUT2** (if used): mean ± SD per construct	Horizontal bars	P4
or	**If you skip 3–4**: method, procedure, results and evidence of the method you used	Follow the 4 headings on the "standard instrument" slide	P3, P4
5	**Issues found**: top 3–5 problems + severity 0–4 + user quotes	Issue table	P3
6	**What you fixed**: before → after	Before / after screenshots	P1

RAW EVIDENCE Keep consent forms, the raw Excel sheet, and photos or video available for the instructor to check

never fabricate data

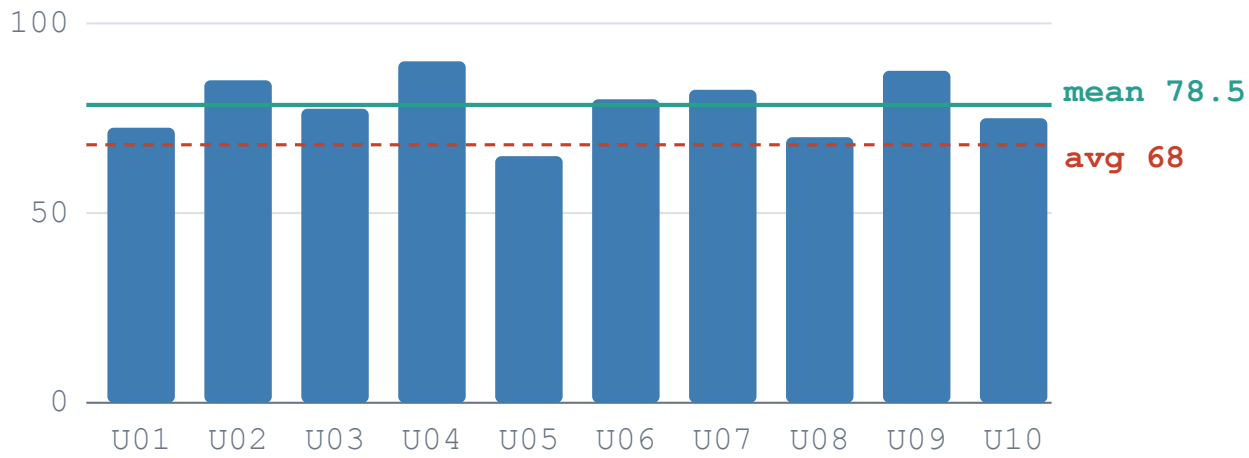
Severity (Nielsen): 0 not a problem
1 cosmetic
2 minor
3 major
4 catastrophe, must fix before release

EXAMPLE RESULTS: CUPLOG

What your results should look like

ILLUSTRATIVE DATA ONLY

SUS per user: **Mean 78.5**, SD 8.0, n = 10, **B+** (above the 68 average)



TAM by construct (Likert 1–5, mean ± SD)

PU	**4.3 ± 0.5**
PEOU	**4.5 ± 0.4**
BI	**4.1 ± 0.6**

With UTAUT2, show it the same way: one bar per construct.

TASK	SUCCESS	MEAN TIME	BEFORE (OLD WAY)	ERRORS / USER
T1 Log one latte	10/10 (100%)	8 s	35 s (paper tally)	0.2
T2 See today's total	10/10 (100%)	5 s	4 min (manual count)	0.0
T3 Fix a wrong entry	7/10 (70%)	21 s	–	1.1

INSIGHT → FIX T3: 3 of 10 users couldn't find the edit button (severity 3) → moved the edit button onto the entry → retested

Always present a number together with **what your team did about it**.

W13 AND W14: FINAL PROJECT PRESENTATION

Final presentations: 11 and 18 Nov 2026

WED 11 NOV (W13)

Teams 1–10

10 minutes per team

WED 18 NOV (W14)

Teams 11–20

10 minutes per team

PORTFOLIO	MUST INCLUDE
P1 Case Study Page	Problem → Process → Outcome
P2 Live product + demo	A working system and a demo video ≤ 3 min
P3 User Evidence Log	Proof that you talked to and tested with real users
P4 Impact Metrics	Before / after + one metric on AI's impact on your team's SDLC
P5 Personal Role Statement	200 words per person, including your AI-use disclosure

TEAM CHECKLIST

What your team must do, by date

OCTOBER

- [ ] **14 Oct** Join the online class
- [ ] **21 Oct** No class; write the Test Plan
- [ ] **Before 28 Oct** Submit the Test Plan
- [ ] **31 Oct** Guest lecture in E4614; bring a laptop

NOVEMBER

- [ ] **4 Nov** Test with real users and collect evidence
- [ ] **11 Nov** Teams 1–10 present the final
- [ ] **18 Nov** Teams 11–20 present the final
- [ ] **22 Nov – 4 Dec** Final exam period

Questions? Comment under the post in Google Classroom